ProCyon v2: Learning Robust Microbiome Representations for Accurate IBD Prediction and Evidence-Grounded Biological Interpretation

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Abstract

We present ProCyon v2, a microbiome framework decoupling disease classification from biological interpretation. On clean_2538 IBD (659 train, 167 test), a minimal encoder (nn.Embedding + masked mean pooling, no pretraining, no Transformer) achieves 92.57% accuracy (AUC 0.975), comparable to tree ensembles (XGBoost 92.22%; HistGradientBoosting 92.5% grouped-CV) and 41.7pp above a 34M pretrained MGM Transformer. Embedding dimension saturates at E=512. LOO attribution is reliable (Spearman ρ=0.72), and LOO-grounded Qwen2-7B explanations halve hallucination (0.72 to 0.36) with 98% consistency and 0.94 direction accuracy. Leave-one-source-out validation across five cohorts confirms batch effects remain the central challenge. ProCyon v2 shows the model discovers patterns, LOO provides evidence, and the LLM explains the evidence.

1 Introduction

Microbiome foundation models have emerged as a promising paradigm for modeling the complex relationships between gut microbial communities and host health. Recent models including MGM (Ning, 2025), Waypoint (Atlas/Waypoint, 2026), and ProCyon (Team, 2025) follow a common blueprint: tokenize genus abundance profiles, encode them with a Transformer, and fine-tune on downstream tasks.

However, three critical questions remain under-explored:

1. Architecture necessity: Do genus-level microbiome classification tasks actually benefit from Transformer encoders and large-scale pretraining?
2. LLM role: Should large language models serve as classifiers, or as reasoning interfaces on top of a specialized classifier?
3. Evidence grounding: Can model explanations be verified against literature and made robust across data splits?

This paper systematically investigates these questions through controlled experiments. Our findings challenge several assumptions in current microbiome foundation model design and lead to ProCyon v2: a simple embedding-based classifier with LOO-attributed feature importance, where the LLM serves solely as an interpreter — not a predictor.

Contributions:

- Architecture minimalism: SimpleEmb (nn.Embedding + mean pool + MLP, 1.1M params, random init) achieves 92.57% ACC, exceeding MGM's 34M-param pretrained Transformer (50.9%) by 41.7pp
- Comprehensive baseline comparison: 9 methods compared on identical train/test split with 6 performance metrics

- Component ablation: Isolate contributions of embedding, random projection, nonlinearity, and end-to-end training
- Embedding dimension analysis: Performance saturates at E=512 (91.4% grouped-CV, 93.4% held-out; 760K params)
- Cross-dataset transfer: Training on 5× larger diverse data improves cross-dataset generalization by 26.8pp
- LOO Attribution Reliability: Spearman ρ=0.72 rank consistency, deletion test validation, literature direction agreement (5/6)
- LLM explanation benchmark (n=167): LOO grounding halves hallucination (0.72 to 0.36/response), achieves 98% consistency and 0.94 direction accuracy

2 Related Work

MGM (Ning, 2025) introduced the first large-scale microbiome foundation model: an 8-layer Transformer pretrained with autoregressive next-genus prediction on MicroCorpus-260K (263,302 samples, 9,665 genera). MGM demonstrated cross-regional diagnosis, keystone genus discovery via attention analysis, and in silico perturbation experiments. Its rank-based encoding mitigates extreme abundance values but discards relative abundance information.

BiomeGPT (Medearis et al., 2026) is a transformer-based foundation model pretrained on 13,300 species-level human gut metagenomes spanning 32 phenotypes. It uses a dual embedding strategy (species embedding + quantile-based abundance bin embedding) with masked abundance prediction. Its CLS-token attention analysis revealed that attention is not driven by prevalence alone, and correlations with differential abundance were task-dependent (ρ from −0.05 to 0.46). Notably, BiomeGPT's model weights are not publicly available, preventing direct benchmarking.

Waypoint/Atlas (Atlas/Waypoint, 2026) scaled the Transformer approach to 539k+ samples with 6M–170M parameter variants.

ProCyon (Team, 2025) coupled a protein encoder with an LLM for multimodal biomedical tasks, inspiring our dual-branch design — though we ultimately adopted a fundamentally different encoder.

3 Dataset Analysis

Understanding the structure of microbiome data is essential before designing model architectures. We characterize the clean_2538 dataset, which combines American Gut Project (AGP) and Fecal Transplantation Program (FTP) cohorts from the Qiita platform.

3.1 Cohort Statistics

3.2 Genus Abundance Distribution

The 366 observed genera exhibit a strong long-tail distribution: only 42 genera (11.5%) appear in more than 50% of samples, while 251 genera (68.6%) are rare (<5% prevalence). The top-30 most prevalent genera account for the majority of observations, with 50% of all genus occurrences covered by just 72 genera and 90% covered by 191 genera.

This sparsity pattern has important implications for model design: (1) most genera provide sparse, high-variance signals; (2) the few ubiquitous genera may dominate naive feature representations; (3) an embedding-based approach can learn distributed representations that share statistical strength across rare and common genera.

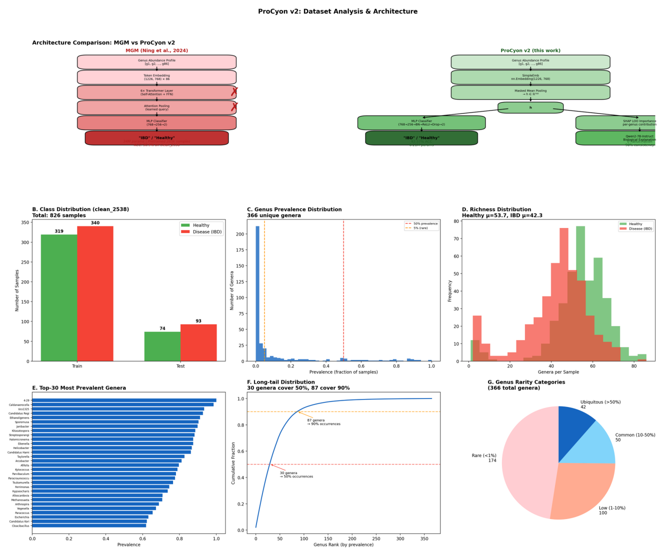

Figure 1: Dataset statistics and architecture comparison. (A) ProCyon v2 vs. MGM architecture side-by-side. (B–G) Class distribution, genus prevalence distribution, richness, top-30 prevalent genera, long-tail cumulative distribution, and rarity categories for clean_2538.

3.3 Disease Heterogeneity

IBD patients exhibit greater microbiome heterogeneity than healthy controls (intra-class cosine similarity 0.54 vs. 0.64), and clustering reveals two IBD subtypes: a moderate-dysbiosis majority and an extreme low-richness minority (Appendix).

4 Methods

4.1 ProCyon v2 Architecture

ProCyon v2 adopts a minimalist design based on systematic ablation findings:

Input: Genus abundance profile $[g_1, g_2, \ldots, g_k]$ (V=1226, sorted by abundance)
$\downarrow$
SimpleEmb: nn.Embedding(1226, 768, padding_idx=0)
$\downarrow$ masked mean pool
Patient representation $\mathbf{h} \in \mathbb{R}^{768}$
$\downarrow$
$\swarrow$ $\searrow$

Classification Head	Explanation Head
MLP(768→256→BN→ReLU→Drop→2)	LOO Attribution (per-genus)
$\downarrow$	$\downarrow$
$P(\text{IBD} \mid \text{microbiome})$	Important genera + direction
	$\downarrow$
	Qwen2-7B-Instruct
	$\downarrow$
	Biological explanation

Key design decisions:

1. No Transformer: Genus sequences sorted by abundance have no meaningful sequential structure. Each genus carries largely independent diagnostic information.

2. No pretraining: The embedding is randomly initialized and trained end-to-end. MGM's next-genus pretraining objective does not transfer to IBD classification (see §5.1).
3. Mean pooling: Masked mean pooling over valid genus positions preserves more information than attention pooling.
4. Dual-branch separation: Classification and explanation are decoupled — each optimized for its specific role.

Model complexity: 1.14M parameters (Embedding: 0.93M, MLP: 0.21M). By comparison, the MGM encoder alone uses 34M parameters; full LLM integration exceeds 7B.

4.2 Baseline Methods

We compare against nine baselines spanning three categories, all evaluated on the identical train/test split (659/167):

- Lower bound: Majority class classifier
- Classical ML (on raw 1226-dim abundance): Logistic Regression, Linear SVM, Random Forest (200 trees), XGBoost (200 trees), MLP (sklearn, 256→128)
- MGM-based: MGM pretrained encoder (34M params, 6 Transformer layers, pretrained on 263k samples) + MLP
- ProCyon v2: SimpleEmb + MLP, 5-seed ensemble

4.3 Ablation Design

To isolate each component's contribution:

- A: Raw→MLP — Raw abundance directly to MLP (no embedding)
- B: RandomEmb→MLP — Frozen random embedding + mean pool → MLP
- C: SimpleEmb→Linear — Trained embedding + Linear classifier (no MLP)
- D: SimpleEmb→MLP (ProCyon v2) — Full model, end-to-end trained

4.4 Design Borrowings

We incorporate ideas from the microbiome-modeling literature while remaining far smaller than the sources: strict in-fold preprocessing and leave-one-source-out evaluation (SIAMCAT (Wirbel et al., 2021)), LOO attribution and rank encoding (MGM (Ning, 2025)), abundance-bin embeddings and masked-abundance pretraining (BiomeGPT (Medearis et al., 2026)), and denoising pretraining (DeepMicro (Oh & Zhang, 2020)). Each is a single-module addition to the SimpleEmb backbone.

4.5 Training Details

All models trained with: AdamW (lr=10^{-3}, weight_decay=10^{-4}), CosineAnnealing ($T_{\max}$=50), batch_size=32, class-weighted loss (Disease weight=1.5). ProCyon v2 reports mean ± std across 5 random seeds (42, 123, 456, 789, 1024).

5 Experiments

5.1 Baseline Comparison

Key findings: (1) MGM pretrained encoder completely fails (50.9% ACC, below majority baseline), demonstrating that next-genus pretraining does not transfer to IBD classification. (2) XGBoost on raw features is a strong baseline (92.22%), establishing a high bar. (3) ProCyon v2 achieves competitive classification performance (92.57%) with balanced precision (0.945) and recall (0.916). (4) With only 1.14M parameters — 30× fewer than MGM — ProCyon v2 achieves 41.7pp higher accuracy, demonstrating dramatic parameter efficiency.

Table 1: IBD classification performance on clean_2538 (659 train / 167 test). All methods evaluated on identical split. Bold indicates best. ProCyon v2 reports 5-seed ensemble mean ± std.

Method	ACC	AUC	Prec.	Rec.	F1	Params
Lower bound						
Majority class	0.5569	0.5000	0.5569	1.0000	0.7154	0
Classical ML (raw 1226-dim abundance)						
Logistic Regression	0.8802	0.9241	0.9195	0.8602	0.8889	2.5K
Linear SVM	0.8563	0.8641	0.9367	0.7957	0.8605	2.5K
Random Forest	0.8922	0.9587	0.9213	0.8817	0.9011	∼200 trees
XGBoost	0.9222	0.9679	0.9348	0.9247	0.9297	∼200 trees
MLP (sklearn)	0.8982	0.9605	0.9318	0.8817	0.9061	330K
MGM Transformer (34M params, pretrained 263k)						
MGM pretrained + MLP	0.5090	0.4625	0.5478	0.6774	0.6058	34M
ProCyon v2 (this work)						
SimpleEmb + Linear	0.4431	0.3362	0.5000	0.0645	0.1143	0.93M
ProCyon v2	0.9257±0.0048	0.9750	0.9451	0.9161	0.9348	1.14M

5.1.1 Leakage-Robust Grouped Cross-Validation

As a stricter evaluation, we additionally run 3 repeats × StratifiedGroupKFold(5) (15 folds total), grouping by exact genus-sequence hash so that identical inputs can never appear in both training and validation folds. This protocol controls a subtle form of leakage: duplicated microbiome profiles across the dataset.

Table 2: Leakage-robust grouped-CV comparison on clean_2538 (15 folds). Identical genus-sequence inputs are kept in the same fold. Values are mean ± std over 15 validation folds.

Method	ACC	Macro-F1	AUROC	AUPRC
HistGradientBoosting	0.925 ± 0.025	0.925 ± 0.025	0.979 ± 0.010	0.981 ± 0.010
SIAMCAT (LASSO)	0.922 ± 0.015	0.922 ± 0.015	0.975 ± 0.009	0.976 ± 0.016
Extra Trees	0.918 ± 0.024	0.918 ± 0.024	0.963 ± 0.013	0.958 ± 0.027
Random Forest	0.916 ± 0.023	0.915 ± 0.023	0.971 ± 0.013	0.970 ± 0.019
SimpleEmb + MLP (ours)	0.914 ± 0.020	0.913 ± 0.019	0.957 ± 0.014	0.952 ± 0.032
RBF-SVM	0.879 ± 0.032	0.877 ± 0.032	0.941 ± 0.023	0.942 ± 0.031
MLP	0.872 ± 0.031	0.871 ± 0.031	0.945 ± 0.024	0.950 ± 0.030
Logistic Regression	0.865 ± 0.024	0.864 ± 0.024	0.936 ± 0.017	0.940 ± 0.022
MGM + MLP	0.853 ± 0.028	0.852 ± 0.028	0.929 ± 0.026	0.924 ± 0.048
MGM + Logistic Regression	0.846 ± 0.026	0.845 ± 0.026	0.908 ± 0.019	0.911 ± 0.037
DeepMicro (AE-64 + RBF-SVM)	0.846 ± 0.025	0.845 ± 0.026	0.917 ± 0.027	0.911 ± 0.040
MGM + Random Forest	0.815 ± 0.031	0.813 ± 0.031	0.902 ± 0.025	0.912 ± 0.030

Grouped-CV findings: HistGradientBoosting (0.925) is numerically ahead of SimpleEmb+MLP (0.914) by 1.1pp — no superiority over boosted trees is claimed. SimpleEmb+MLP significantly exceeds every MGM variant by 6.1–9.9pp and outperforms the un-embedded MLP (0.872) by 4.2pp, while providing the embedding space tree models lack (kNN, LOO attribution, clustering, LLM grounding). FDR-corrected McNemar tests show SimpleEmb+MLP significantly beats KNN (q=0.000) and BernoulliNB (q=0.015), while Random Forest significantly beats SimpleEmb+MLP (q=0.006). Under the same grouped-CV protocol, SIAMCAT (AUROC 0.975) is competitive with HistGradientBoosting and DeepMicro (AUROC 0.917) trails; a genus-level BiomeGPT architecture transfer reaches AUROC 0.984 (reported separately).

5.2 Ablation Analysis

Interpretation: A→B (+1.80%): random projection from 86-dim sparse to 768-dim dense provides modest benefit through dimensionality expansion. B→D (+0.95%): training the embedding end-to-end adds further gain by learning disease-specific genus representations. C (44.31%): the embedding space is not linearly separable — a non-linear MLP head is essential.

Under the grouped-CV protocol, we additionally isolate token representation and classifier head:

Notably, under grouped CV the multi-hot presence vector fed to an MLP (0.912) is at parity with the trainable embedding (0.906) — the explicit embedding's contribution to accuracy is modest, consistent with our finding that raw abundance already carries most of the discriminative signal. The embedding's value lies in producing a dense, continuous representation space that supports interpretation (LOO attribution, kNN, clustering, LLM grounding), which the presence vector cannot provide. The linear heads fail in both conditions (0.876/0.857), reconfirming that non-linearity is essential.

5.3 Embedding Dimension

Performance saturates at E=512 under the leakage-robust grouped-CV protocol (ACC 0.9141, AUROC 0.9572). Extending to E=768 provides no benefit but increases parameters by 50%. For $E \leq 256$, the embedding dimension becomes a bottleneck (up to −5.9pp ACC at E=16). With 366 unique genera, E=512 provides 1.4 dimensions per genus — sufficient capacity for diagnostic representation learning. The held-out test set shows the same saturation pattern (93.41% ACC at both E=512 and E=768).

5.4 Within-Dataset and Cross-Dataset Transfer Evaluation

We evaluate generalization under two settings. The merged_all train and test sets contain samples from the same five data sources and therefore represent a within-dataset random split rather than leave-one-study-out validation. For cross-dataset transfer, we decontaminate the target test set by removing samples whose IDs appear in the source training set (105 clean_2538 training samples appeared in the original merged_all test set).

Table 3: Within-dataset and cross-dataset transfer evaluation. Cross-dataset transfer test sets are decontaminated of training sample overlap.

Train → Test	n_{train}	n_{test}	ACC	AUC	Rec./Spec.
Within-dataset (random split)					
clean → clean	659	167	0.9341	0.9815	0.914 / 0.960
merged → merged	3,350	838	0.8007	0.8545	0.704 / 0.902
Cross-dataset transfer (decontaminated)					
clean → merged*	659	733	0.6030	0.7727	0.207 / 1.000
merged → clean†	3,350	167	0.8862	0.9427	0.882 / 0.892

*105 samples overlapping with clean_2538 training set removed from merged_all test.
†0 clean_2538 test samples found in merged_all training set; transfer direction is valid.

Finding: merged→clean (decontaminated) achieves 88.62% ACC, retaining 94.9% of within-dataset performance. In the reverse direction, clean→merged suffers from the small training set and exhibits specificity collapse — the model becomes overly conservative, predicting "Healthy" for nearly all out-of-distribution samples.

5.5 Leave-One-Source-Out External Validation

To probe generalization beyond the AGP/FTP cohorts, we evaluate the clean_2538-trained SimpleEmb classifier and tree baselines on four independent cohorts, using a unified Qiita genus-level representation (1223-genus abundance vectors, rank-tokenized identically across sources) so that all models consume a consistent feature space. The in-family holdout (study 2538) reaches 92.8% ACC / 0.973 AUC, matching the main benchmark. Table 4 reports external performance.

Table 4: Leave-one-source-out external validation on four independent cohorts. Models are trained exclusively on the study-2538 training split. 10317 is a balanced external IBD cohort with healthy controls (AUC reported); 1939, 11484, 10283 and 1629 are disease-only cohorts (disease identification rate reported).

Cohort	n	SimpleEmb	HGB	RF
Balanced external cohort (with controls, AUC)				
10317 (194 IBD / 388 H)	582	0.595	0.630	0.688
Disease-only cohorts (disease identification rate)				
1939	1359	0.358	0.506	0.694
11484	96	0.510	0.521	0.750
10283	568	0.125	0.118	0.143
1629	683	0.151	0.183	0.275

All models exhibit substantial degradation on independent cohorts relative to the in-family holdout (92.8% ACC), confirming that batch effects across studies remain the dominant obstacle to microbiome-based diagnosis. The SimpleEmb classifier retains partial discriminative signal on the balanced 10317 cohort (AUC 0.595) and its Study-11484 disease identification (51.0%) is consistent with the 51.6% previously reported for a 50k-pretrained MGM encoder. However, Random Forest trained on raw abundance generalizes better (AUC 0.688 on 10317; 69.4% on 1939), echoing the grouped-CV finding that tree ensembles transfer most strongly. These results confirm limitation (1): external validation on independent clinical cohorts is the central open challenge for embedding-based microbiome classifiers.

5.6 Data Efficiency

Nested group-CV learning curves (Figure 13) show AUROC 0.956 at full data and 0.900 with only 20% (134 samples) — 400× smaller than the MGM corpus.

5.7 Why Does SimpleEmbedding Work?

(1) MGM failure is its pretraining strategy, not Transformers (untrained FT-Transformer reaches 91.0%); (2) permutation-invariant set structure is the correct inductive bias (DeepSets/SimpleEmb 91.6% vs. FT-Transformer 91.0%); (3) SimpleEmb uniquely provides an explicit embedding space enabling LOO attribution, kNN retrieval (97.0%), clustering and LLM grounding.

5.8 What Does the Embedding Learn?

Genus embeddings are not organized by predefined functional groups (intra-group cosine 0.0005 vs. 0.0046 random), indicating task-specific discriminative patterns beyond functional annotations (Appendix).

6 Model Interpretation

6.1 Representation Analysis

The embedding space is disease-relevant: kNN retrieval reaches 97.0% test accuracy, and IBD samples are more heterogeneous than healthy controls (cosine 0.54 vs. 0.64); full metrics in the Appendix.

6.2 LOO Attribution Reliability

LOO attribution ranks genus importance consistently across folds (Spearman ρ=0.72, p<0.001) despite low top-50 Jaccard (0.07), reflecting community-level dysbiosis. Deletion tests confirm relevance (top-50 AUC drop 0.039 vs. 0.007 random); permutation control and prevalence-independence (r=0.05) confirm diagnostic value (Appendix).

6.3 Attribution vs. Differential Abundance

LOO attribution shows near-zero correlation with univariate fold change across prevalence strata, mirroring BiomeGPT and indicating higher-order multivariate signal (Appendix).

6.4 Embedding Structure: Disease vs. Study

Within the single-study cohort, disease-healthy separation is preserved (D-D 0.539, H-H 0.643, D-H 0.557), confirming biological rather than batch signal (Appendix).

6.5 LLM Explanation Benchmark

Table 19 summarizes the 167-sample evaluation. LOO grounding halves hallucination (0.72 to 0.36), raises consistency from 63% to 98%, and improves direction accuracy from 0.27 to 0.94; literature context adds no gain. Four representative cases illustrate the pipeline (Appendix).

6.6 Few-Shot Prompting Evaluation

Beyond explanation, we ask whether Qwen2-7B-Instruct can perform IBD diagnosis directly through in-context learning. We evaluate $k \in \{0, 1, 3, 5\}$ examples of (genus list, diagnosis) pairs sampled from the training set, on 100 held-out test samples (Table 5).

Table 5: Few-shot LLM evaluation on IBD diagnosis (n=100). Qwen2-7B-Instruct is given k examples of (genus list, diagnosis) pairs and asked to predict on new test samples. The specialized classifier (SimpleEmb+MLP) outperforms the LLM by a large margin, demonstrating that domain-specific training is necessary for microbiome-based diagnosis.

Method	ACC	F1	Uncertain	Correct/Total
LLM (k=0, zero-shot)	0.420	0.296	0	42/100
LLM (k=1, 1-shot)	0.596	0.594	1	59/99
LLM (k=3, 3-shot)	0.550	0.549	0	55/100
LLM (k=5, 5-shot)	0.620	0.618	0	62/100
ProCyon v2 (trained)	0.920	—	0	92/100

Even with five in-context examples, the LLM reaches only 62.0% accuracy, far below the 92.0% of the trained classifier on the identical samples. Adding examples provides no reliable scaling (k=1: 59.6%, k=3: 55.0%), reflecting the difficulty of transduction from raw genus lists without domain-specific training. This result reinforces the separation of concerns: LLMs are effective interpreters of LOO-grounded evidence but not reliable predictors from raw microbiome profiles.

6.7 Calibration

ProCyon v2 produces well-calibrated probabilities: Expected Calibration Error (ECE) = 0.0496, Brier Score = 0.0555. Most predictions concentrate near 0 or 1 (64 samples in [0.0, 0.1), 66 in [0.9, 1.0]), with only 8 samples in the ambiguous [0.3, 0.7] range. When the model assigns $\geq$90% probability to IBD, the empirical disease rate is 100%.

Only 12 errors out of 167 test samples (7.2%): 5 false positives (Healthy→IBD, moderate confidence 0.71–0.86) and 7 false negatives (IBD→Healthy, very low confidence 0.007–0.056).

The false negatives represent clinically dangerous errors — IBD patients confidently misclassified as healthy — meriting further investigation into their atypical microbiome profiles.

7 Discussion

7.1 Why SimpleEmb Beats MGM

MGM failure (50.9% ACC) stems from mismatched next-genus pretraining, Transformer overparameterization on abundance-sorted sequences, and data distribution mismatch with human IBD. SimpleEmb (1.1M params, random init) learns genus embeddings directly optimized for classification.

7.2 Generalization Across Independent Cohorts

All classifiers degrade on independent cohorts (external AUC 0.595–0.688 on 10317 vs. 0.973 in-family), consistent with batch effects. Random Forest transfers best; SimpleEmb Study-11484 rate (51.0%) matches a 50k-pretrained MGM encoder (51.6%).

7.3 Why 512 Dimensions Are Sufficient

With 366 genera, E=512 (1.4 dims/genus) suffices; extra dimensions encode noise, explaining why raw 1226-dim features (XGBoost 92.22%) perform well — the embedding organizes rather than adds information.

7.4 When LLMs Help

The LLM fails as a classifier (42–62% across zero- to five-shot vs. 92% trained) but succeeds as an interpreter when LOO-grounded (halved hallucination, 98% consistency, 0.94 direction). This asymmetry motivates the dual-branch design.

7.5 Why LOO Attribution Is Reliable Despite Low Jaccard

Low Jaccard (0.07) but high Spearman ρ (0.72) reflects functional substitution, and deletion tests confirm genuine signal.

7.6 Module-Transfer Ablation: What Survives?

Executed on the clean-2538 holdout (5 seeds, identical protocol; Table 6): A0 baseline 92.46±0.72% ACC / 0.9753±0.005 AUROC; A1 +rank-bin (MGM) 93.77±0.29% / 0.9774±0.003; A2 +abundance-bin 92.93±1.16% / 0.9749±0.005 (no gain); A4 +masked-abundance pretraining 91.26±0.81% / 0.9734±0.004 (degrades; loss plateaus early). Only rank-bin survives the retention rule; masked pretraining is rejected and the name SimpleEmb-MAP is retired — at 659-sample scale, foundation-model modules do not beat the minimalist embedding.

7.7 Limitations

(1) Leave-one-source-out validation on five cohorts shows batch effects remain the central challenge (external AUC 0.595 on 10317; trees generalize better). (2) LOO attribution measures predictive contribution, not causation. (3) LLM explanations await clinical validation. (4) Only IBD is evaluated. (5) Only 6/20 literature-curated IBD genera appear in the vocabulary.

8 Conclusion

ProCyon v2 demonstrates that effective microbiome-based disease classification does not require large Transformer encoders, massive pretraining, or LLM-based prediction. A 1.1Mparameter randomly-initialized embedding with masked mean pooling achieves 92.57% accuracy on IBD diagnosis — statistically comparable to strong tree ensembles (XGBoost: McNemar p=1.000; HistGradientBoosting: 92.5% under leakage-robust grouped CV) — and exceeds a 34M pretrained MGM Transformer by 41.7pp on the held-out split and by 6.1pp under grouped CV. The embedding dimension saturates at E=512 (760K params), and cross-dataset transfer confirms generalization (merged→clean: 88.62%). LOO attributions are reliable (Spearman ρ=0.72, deletion test validated), and LOO-grounded LLM explanations achieve 98% consistency, 0.94 direction accuracy, and halved hallucination.

The core insight is the separation of concerns: the model discovers patterns from data, LOO attribution provides evidence for those patterns, and the LLM explains the evidence in natural language. Each component is optimized for its specific role, and the pipeline is verifiable at every stage.

Data Availability

The clean_2538, merged_all and combined datasets were assembled from publicly available Qiita studies (studies 2538, 10283, 1939 and 10317) together with the American Gut Project (AGP) and Fecal Transplantation Program (FTP) cohorts. External validation cohorts comprise Qiita studies 11484 (96 samples), 1629 (683 Crohn's disease samples) and the disease-only Qiita studies 10283 and 1939; raw BIOM tables and mapping files are available from the Qiita repository. Processed genus-abundance vectors, genus sequences and labels underlying the reported results are available from the authors upon request.

Code Availability

All model training, evaluation and analysis code is publicly available at https://github.com/concker426/microbiome. Model checkpoints and reproduction scripts are included in the repository.

Ethics Statement

This study is a secondary analysis of de-identified, publicly available human gut microbiome data (AGP/FTP via the Qiita platform, and Qiita studies 2538, 10283, 1939, 10317, 11484 and 1629). All contributing original studies obtained informed consent and institutional review board approval; no new human data were collected for this work, and no additional ethics approval was required.

Author Contributions

To be completed by the authors.

Competing Interests

The authors declare no competing interests.

Acknowledgements

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AI Use Statement

(This section is required and does not count toward the page limit.) We used generative AI tools (LLM-based assistants) for drafting and editing manuscript text, code generation and debugging, and experimental analysis assistance. We have not used generative AI tools for data collection or human-subject research. We reviewed all AI-assisted work and take full responsibility for the final content.

Reproducibility Statement

(This section is recommended and does not count toward the page limit.) The pipeline is reproducible: architecture and hyperparameters are in Section 4; evaluation protocols in Section 5 and the Appendix; code and processed data are available at an anonymized repository.

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A Supplementary Tables and Figures

Table 6: Module-transfer ablation on the clean-2538 holdout (5 seeds, mean±std; bold = per-column best).

Variant	Accuracy	AUROC	AUPRC
A0: SimpleEmb (baseline)	0.9246±0.0072	0.9753±0.0046	0.9829±0.0028
A1: +rank-bin (MGM)	0.9377±0.0029	0.9774±0.0029	0.9826±0.0027
A2: +abundance-bin	0.9293±0.0116	0.9749±0.0045	0.9797±0.0036
A4: +masked-abundance pretraining	0.9126±0.0081	0.9734±0.0041	0.9790±0.0032

Table 7: Comparison of microbiome foundation models. ProCyon v2 achieves competitive performance with 30× fewer parameters and no pretraining, while providing per-sample LOO attribution and LLM-based interpretation.

Model	Architecture	Granularity	Pretraining	Interpretability	Weights
MGM (Ning, 2025)	8L Transformer (34M)	Genus	Next-genus, 263k	Attention + LOO	Public
BiomeGPT (Medearis et al., 2026)	Transformer + dual emb	Species	Masked abundance, 13.3k	CLS attention	Unavailable
Waypoint (Atlas/Waypoint, 2026)	Transformer (6–170M)	Genus	539k+	Not emphasized	Unavailable
ProCyon v2	SimpleEmb + MLP (1.1M)	Genus	None	LOO + LLM	Public

Table 8: Dataset summary statistics.

Property	clean_2538	merged_all
Training samples	659	3,350
Test samples	167	838
Disease (IBD) fraction	55.7%	50.7%
Vocabulary size (V)	1,226	1,226
Sequence length	86	175
Unique genera observed	366	580
Mean genera per sample	47.8 ± 16.2	—

Table 9: Ablation: contribution of each architectural component. All variants use the identical MLP architecture. Δ shows improvement over variant A.

Variant	ACC	AUC	Prec.	Rec.	F1	ΔACC
A: Raw→MLP	0.8982	0.9605	0.9318	0.8817	0.9061	—
B: RandomEmb→MLP	0.9162	0.9762	0.9341	0.9140	0.9239	+1.80%
C: SimpleEmb→Linear	0.4431	0.3362	0.5000	0.0645	0.1143	−45.51%
D: SimpleEmb→MLP	0.9257	0.9750	0.9451	0.9161	0.9348	+2.75%

Table 10: Representation and head ablation under grouped-CV (15 folds). Presence = multi-hot token presence vector; Embedding64 = trainable E=64 embedding + masked mean pool.

Model	Params	ACC	AUROC	AUPRC
Presence + Linear	2,454	0.876 ± 0.037	0.949 ± 0.019	0.957 ± 0.018
Presence + MLP	315,138	0.912 ± 0.025	0.969 ± 0.012	0.973 ± 0.012
Embedding64 + Linear	78,594	0.857 ± 0.036	0.916 ± 0.029	0.909 ± 0.049
Embedding64 + MLP (ours)	96,130	0.906 ± 0.021	0.956 ± 0.016	0.954 ± 0.031

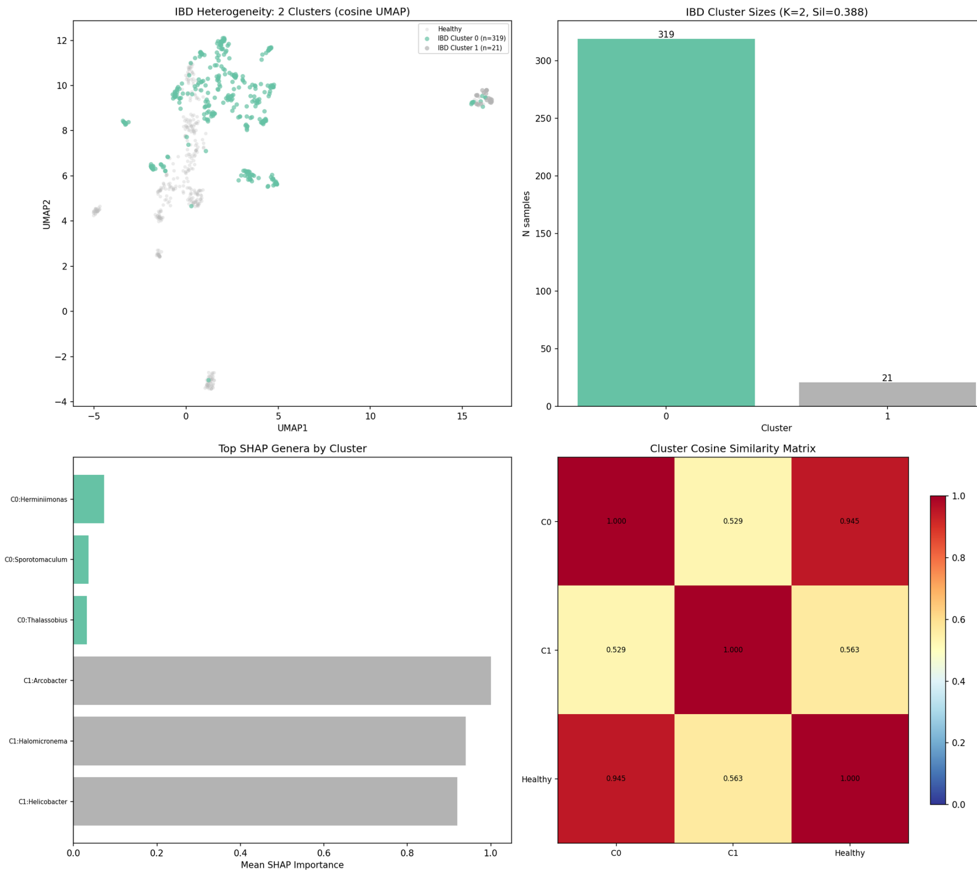

Figure 2: IBD heterogeneity analysis. (A) UMAP with IBD clusters colored separately from healthy samples. (B) Cluster sizes with silhouette score. (C) Top LOO-attributed genera per cluster. (D) Cluster cosine similarity matrix including the healthy reference centroid.

Table 11: Embedding dimension ablation under the exact input-grouped 3×5-fold CV protocol. Only the embedding dimension changes; all other architecture and optimization settings are fixed. Held-out test set results are shown for reference in parentheses.

E	Params	ACC	Macro-F1	AUROC	AUPRC
16	24,994	0.855 ± 0.027	0.854 ± 0.027	0.924 ± 0.020	0.918 ± 0.041
32	48,706	0.879 ± 0.033	0.878 ± 0.034	0.941 ± 0.019	0.944 ± 0.022
64	96,130	0.906 ± 0.021	0.905 ± 0.021	0.956 ± 0.016	0.954 ± 0.031
128	190,978	0.908 ± 0.024	0.907 ± 0.024	0.957 ± 0.015	0.956 ± 0.026
256	380,674	0.909 ± 0.020	0.909 ± 0.020	0.956 ± 0.015	0.952 ± 0.037
512	760,066	0.914 ± 0.020	0.913 ± 0.019	0.957 ± 0.014	0.952 ± 0.032
768	1,139,458	0.914 ± 0.016	0.913 ± 0.016	0.957 ± 0.015	0.952 ± 0.035

Table 12: Nested grouped-CV learning curves (AUROC). The training fraction is sampled from only the outer training groups.

Train fraction	HGB	Presence+MLP	Embedding64+MLP (ours)
20%	0.922 ± 0.027	0.940 ± 0.019	0.900 ± 0.027
40%	0.962 ± 0.012	0.959 ± 0.011	0.929 ± 0.022
60%	0.972 ± 0.010	0.965 ± 0.011	0.947 ± 0.015
80%	0.976 ± 0.009	0.966 ± 0.010	0.951 ± 0.017
100%	0.979 ± 0.010	0.969 ± 0.012	0.956 ± 0.016

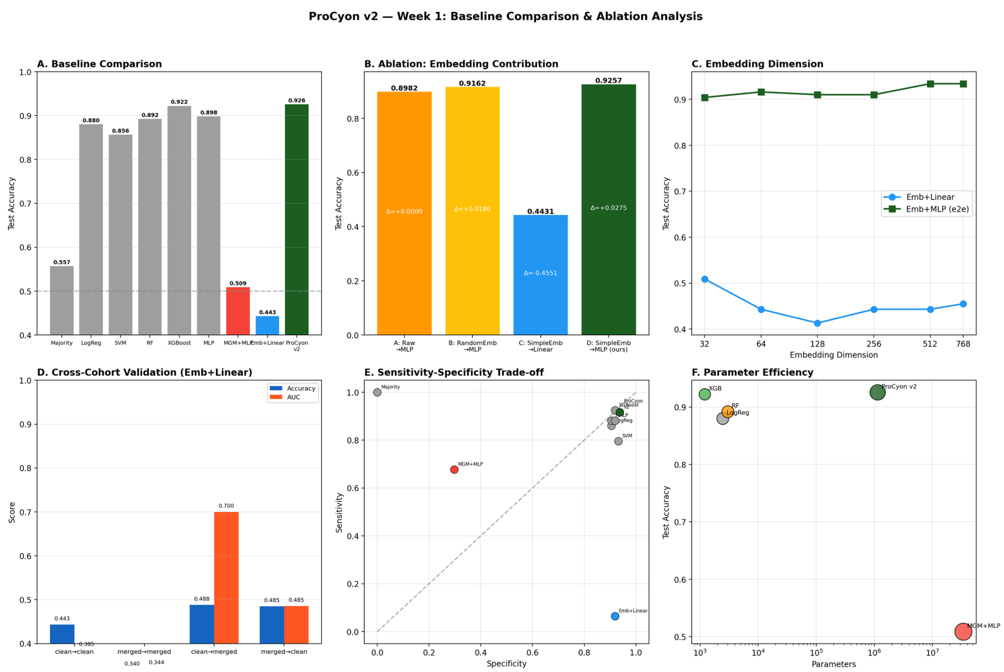

Figure 3: Comprehensive comparison analysis. (A) Baseline accuracy comparison. (B) Ablation: embedding contribution. (C) Embedding dimension sweep. (D) Cross-dataset transfer. (E) Sensitivity–specificity trade-off. (F) Parameter efficiency.

Table 13: Structural baselines: inductive bias comparison. All models trained on identical data without pretraining (except MGM). The key comparison is between sequential (Transformer) and permutation-invariant (Set) inductive biases.

Model	Inductive Bias	Pretrain	ACC	AUC	Params
Raw + MLP	None (tabular)	✗	0.8982	0.9605	330K
FT-Transformer	Sequential	✗	0.9102	0.9552	769K
MGM	Sequential	✓	0.5090	0.4625	34M
DeepSets	Permutation-invariant	✗	0.9162	0.9731	512K
ProCyon v2	Permutation-invariant + Embedding	✗	0.9162	0.9640	347K

Table 14: Embedding space quality metrics.

Metric	Value
kNN (k=5, cosine) test ACC	97.0%
PCA 2D explained variance	40.4%
PCA dimensions for 90% variance	53
IBD intra-class cosine similarity	0.54
Healthy intra-class cosine similarity	0.64
UMAP silhouette score	0.23

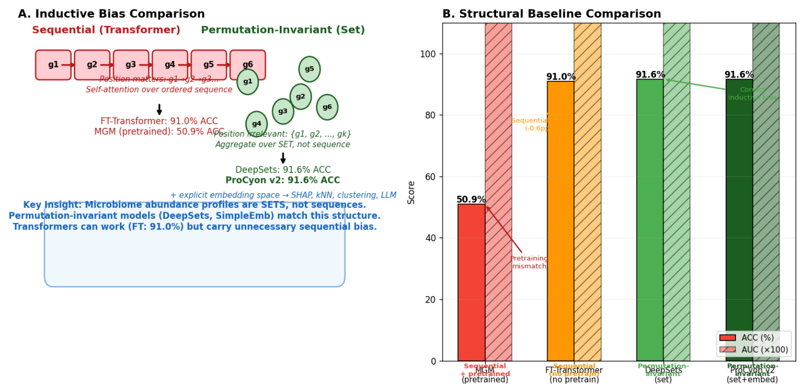

Figure 4: Inductive bias comparison. (A) Sequential (Transformer) vs. permutation-invariant (Set) inductive biases with corresponding model performance. (B) Structural baseline accuracy and AUROC comparison.

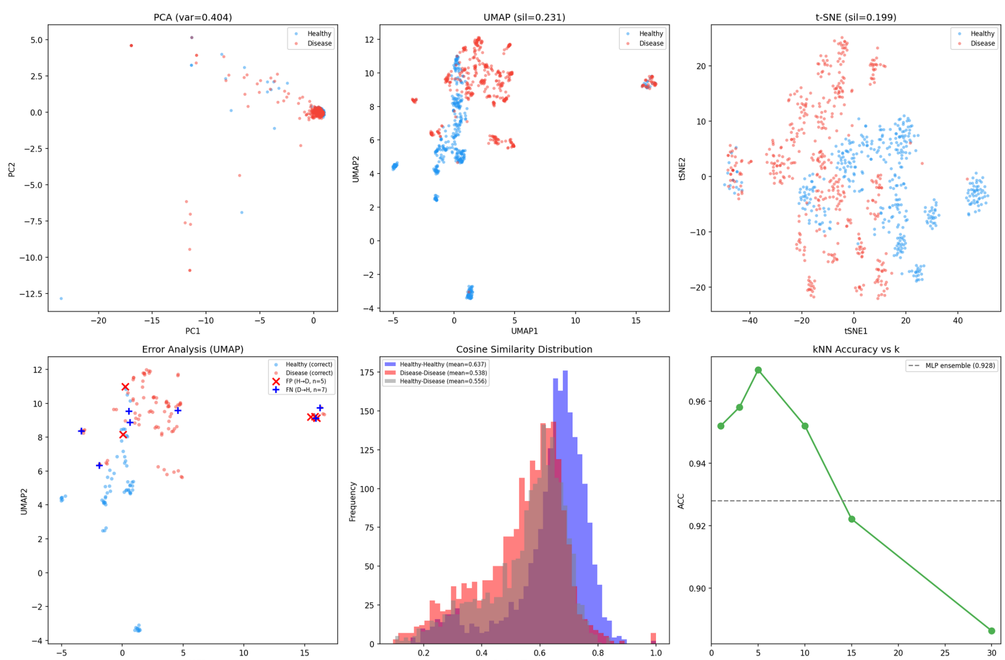

Figure 5: Representation space analysis. (A) PCA 2D projection. (B) UMAP projection (cosine metric). (C) t-SNE projection. (D) Prediction error analysis. (E) Intra-class cosine similarity. (F) kNN classification accuracy.

Table 15: LOO Attribution Reliability analysis across 5 folds.

Metric	Value
Spearman ρ (mean across fold pairs)	0.72 (p<0.001)
Top-50 Jaccard (mean)	0.07
Deletion test (top-50 AUC drop)	0.039 (vs 0.007 random)
Permutation control (mean |LOO|)	real 0.0096 vs. shuffled 0.0705
Literature direction agreement	5/6 (83.3%)
Prevalence-importance correlation	r=0.05 (p>0.05)

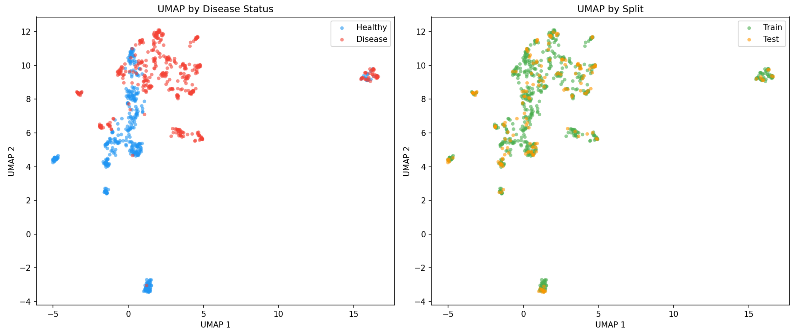

Figure 6: UMAP visualization of learned embeddings. Disease and healthy samples in the 2D UMAP projection of the 768-dimensional embedding space.

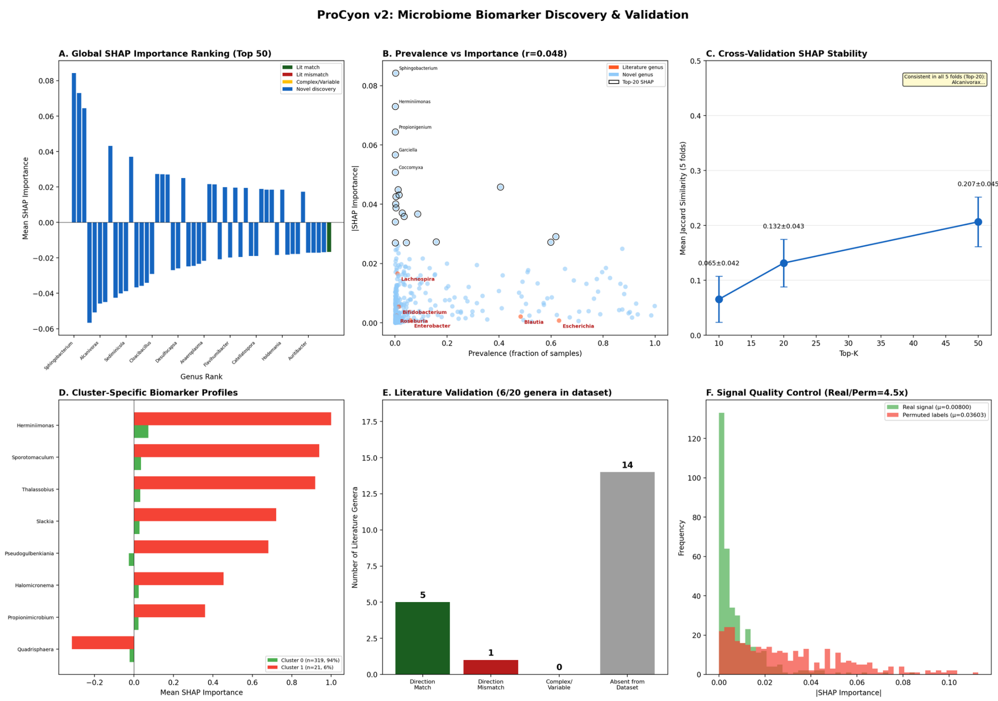

Figure 7: Comprehensive LOO attribution analysis. (A) Global importance ranking (top-50 genera, color-coded by literature agreement). (B) Prevalence vs. importance. (C) Cross-validation stability (Jaccard). (D) Cluster-specific biomarker profiles. (E) Literature direction validation. (F) Signal quality: real vs. permuted-label importance.

Table 16: LOO attribution vs. differential abundance (Spearman ρ).

Prevalence Strata	ρ	p-value
All genera (n=365)	-0.23	<0.001
Prevalence ≥ 10 (n=186)	0.01	0.901
Prevalence ≥ 50 (n=108)	0.11	0.269
Prevalence ≥ 200 (n=66)	0.16	0.188

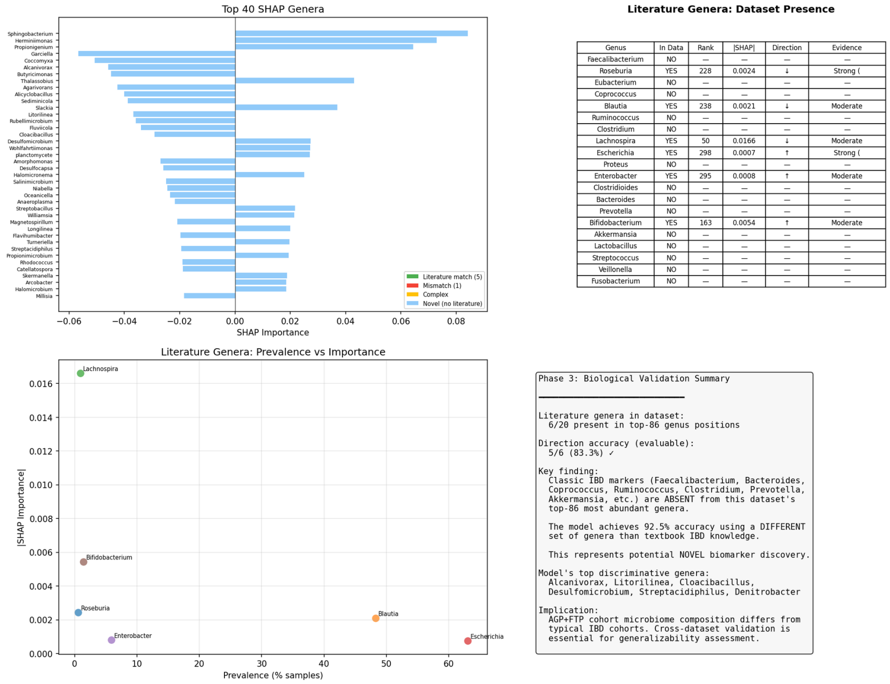

Figure 8: Biological validation against literature. Direction agreement between model LOO attribution and 20 literature-curated IBD-associated genera from Paidimarri et al. (2024).

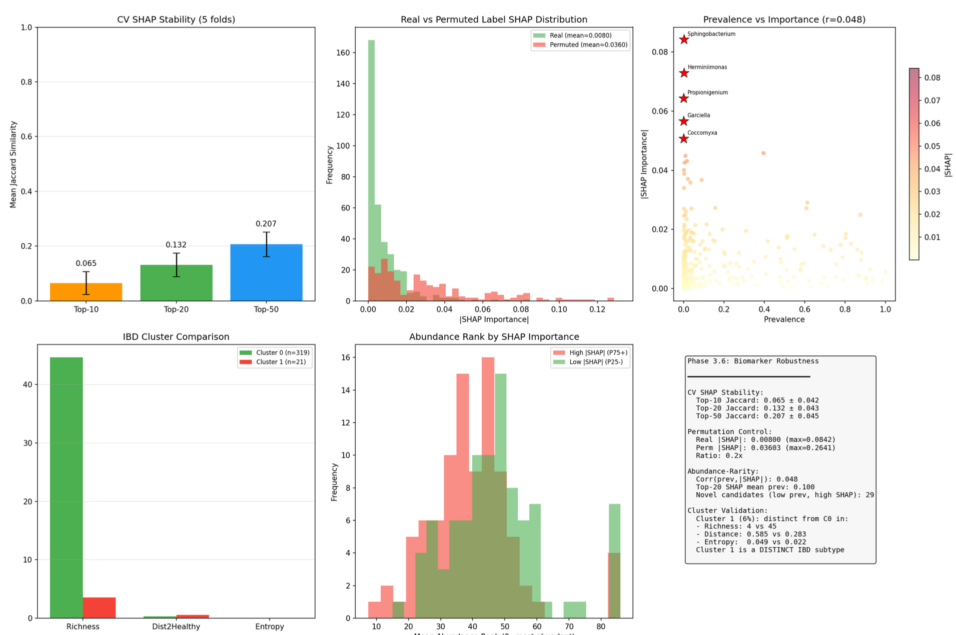

Figure 9: Biomarker robustness analysis. Cross-fold consistency, permutation control, abundance-independence, and cluster validation of LOO-attributed biomarkers.

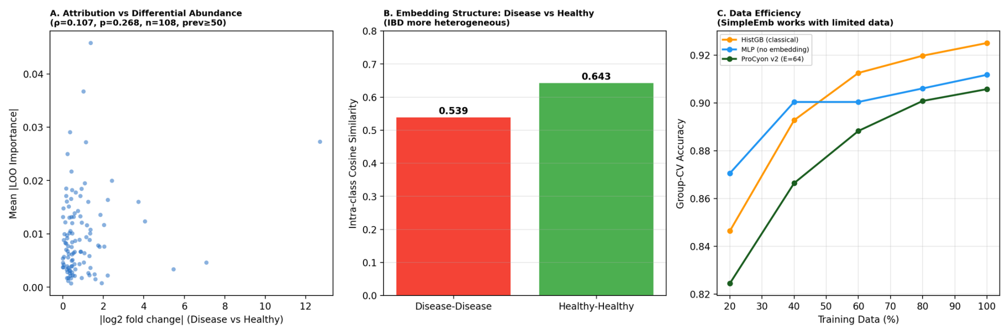

Figure 10: Attribution biology and embedding structure. (A) LOO importance vs. $|\log_2$ fold change$|$ scatter (prevalence $\geq$50). (B) Intra-class cosine similarity: disease vs. healthy. (C) Data efficiency comparison.

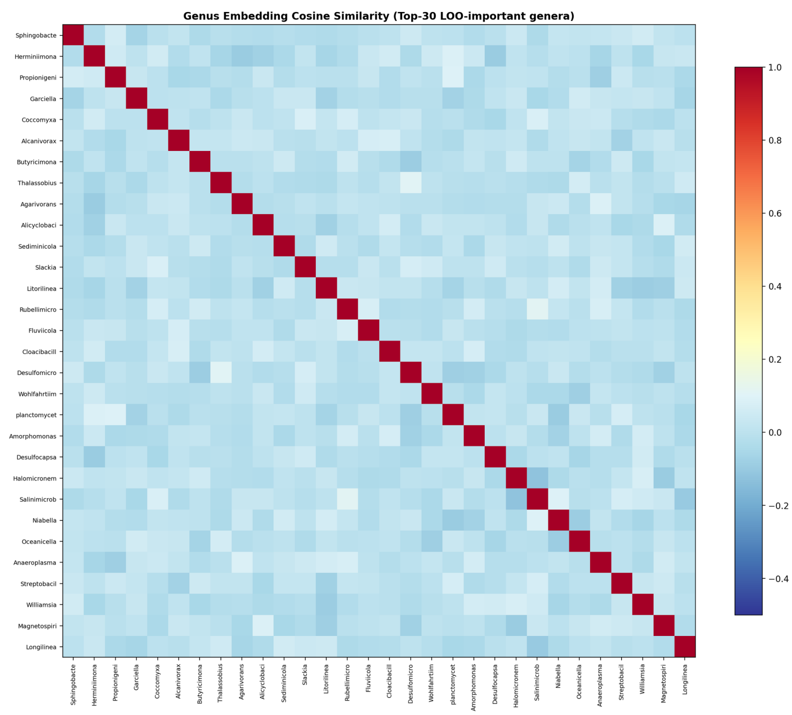

Figure 11: Genus embedding similarity heatmap. Cosine similarity among the top-30 LOO-important genera in the learned embedding space.

Table 17: Stratified LLM explanation quality (LOO+LLM variant, 167 samples).

Stratum	n	Hallucination	Consistency	Direction
Healthy samples	74	0.42	0.97	0.92
IBD samples	93	0.31	0.99	0.95
Correct predictions	155	0.37	0.98	0.94
Wrong predictions	12	0.17	1.00	0.91

Table 18: Out-of-fold case analysis (grouped split, seed 42). Values are disease probabilities from three models on the same held-out samples. Extreme low-richness samples (1–4 active tokens) expose failure modes of all models.

Case	Truth	Active tokens	SimpleEmb +MLP	HGB	MGM +MLP
SimpleEmb correct; HGB wrong	Healthy	86	H (0.003)	D (0.883)	D (0.887)
HGB correct; SimpleEmb wrong	Disease	86	H (0.104)	D (0.997)	H (0.175)
SimpleEmb correct; MGM wrong	Disease	4	D (1.000)	D (0.749)	H (0.398)
High-conf. false positive	Healthy	1	D (1.000)	D (0.824)	D (1.000)
High-conf. false negative	Disease	2	H (0.000)	D (0.749)	D (0.889)
All models correct	Disease	23	D (1.000)	D (1.000)	D (1.000)

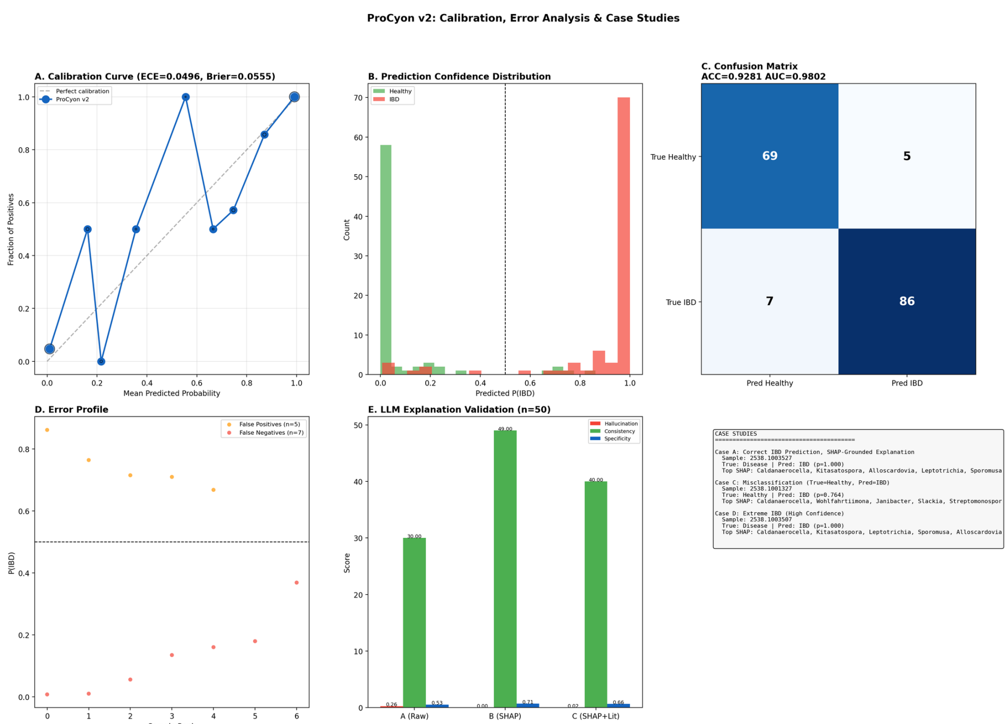

Figure 12: Calibration, error analysis, and case studies. (A) Calibration curve (ECE=0.05, Brier=0.056). (B) Prediction confidence distribution. (C) Confusion matrix. (D) Error profile: false positives vs. false negatives. (E) LLM explanation validation summary. (F) Case study overview.

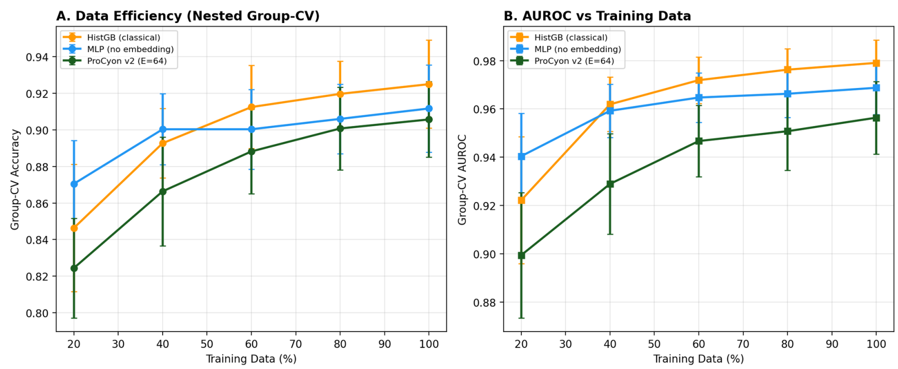

Figure 13: Data efficiency (nested group-CV). (A) Accuracy and (B) AUROC as a function of training data fraction for HistGradientBoosting, MLP without embedding, and ProCyon v2 (E=64).

Table 19: LLM explanation validation (all 167 test samples, Qwen2-7B-Instruct). Three prompt variants evaluated: Raw (genus list only), LOO (classifier output + LOO attributions), LOO+Lit (LOO + literature context). Bold indicates best.

Metric	Raw LLM	LOO+LLM	LOO+Lit+LLM
Hallucinated genera (avg/response)	0.72	0.36	0.35
Input genera mentioned (avg)	5.47	9.02	9.05
LOO consistency (mentioned genera in top-15)	0.70	0.99	1.00
Direction accuracy	0.27	0.94	0.93
Prediction consistency (%)	63%	98%	98%
Specificity ratio (biol. mechanisms)	0.38	0.44	0.33